Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/27/2004
PI Name:	Valles, James M. Ph.D.		
Project Title:	Magnetic Field Gradient Levitation System For Physics and Biophysics		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Biological physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	former web page (chgd 4/07): http://www.physics.brown.edu/		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/01/2001	**End Date:**	04/29/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth"s gravity		

We are developing a Magnetic Field Gradient Levitation (MFGL) apparatus as a ground based system for simulating a low or variable gravity environment for samples made of diamagnetic materials. The system design takes advantage of recent technological advances that have made it possible to create a superconducting solenoid with a room temperature bore that generates a magnetic force strong enough to levitate or cancel the effects of gravity in common organic materials (e.g. water, proteins, polypropylene, etc.). The apparatus will be conducive for pre-space flight studies of the sources of gravitational sensitivity in biological systems and for the investigation of soft condensed matter systems in the virtual absence of gravity. Initial experimental tests of the apparatus will be performed on two biological systems, frog embryos of the species, Xenopus laevis and paramecium. These systems have shown well-defined gravitational sensitivity in experiments on the Space Shuttle. The technique of Magnetic Field Gradient Levitation (MFGL) is being

Task Description:	applied successfully in low temperature studies of fundamental physics in pure superfluid helium and in tests of potential pure superfluid helium experiments for space flight. The primary objective of this project is to extend the MFGL technique to studies of organic physical and biological systems that are composed of more than one type of material and exist in a room temperature environment. We will work to determine the efficacy of MFGL as a means of simulating low gravity on the ground for biological systems and as a means of screening potential systems for experiments in space. In addition, we will endeavor to develop insight into the potential of methods for manipulating systems in the ultra low gravity environment of space using magnetic forces. We take our ability to know up from down for granted relying on the force of gravity to provide the cues. How we, as biological systems, sense and convert these cues into a biological response is important to understand. For example, it is of practical interest to know how prolonged exposure to the low gravity environment of space leads to the documented adverse effects on the immune and skeletal systems of astronauts. In principle, careful biology experiments during spaceflight can reveal the answers. Knowing which experiments to do, however, often requires a great deal of spade work that is impractical to carry out in space. Consequently, experimental methods for simulating low gravity conditions on earth are very desirable. This work will determine the efficacy of Magnetic Field Gradient Levitation (MFGL) as such a method. It is also anticipated that MFGL will be useful for screening other potential space shuttle experiments in physics and biology and may lead to the development of novel magnetic methods of manipulating objects in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A series of experiments were performed in our laboratory and the National High Magnetic Field Laboratory in Tallahassee to obtain video of Paramecia swimming under the influence of strong magnetic body forces and magnetic torques. Magnetic body forces equivalent to subjecting Paramecia to centrifugation at up to 4.5 g's and equivalent to inverting gravity up to 2.5 g were applied. Videotapes of large populations swimming in the different force environments were obtained and are under analysis. Qualitatively, the swimming speeds with and against the magnetic body forces change in the same manner as they do in response to centrifugation. Magnetic torques have been discovered to align swimming Paramecia. A detailed analysis of the changes in their swimming trajectories is underway to assess whether the response of the Paramecia to magnetic fields is purely passive or contains an active component. As a senior thesis project, preliminary experiments on a molecular system known to exhibit gravitational sensitivity were performed. These involved the purfication and polymerization of tubulin in earth's gravity. A paper describing the use of magnetic forces to vary the effective buoyancy of cells was submitted and accepted for publication in Applied Physics Lettters. The PI gave six invited talks (in Tokyo, at the National High Magnetic Field Laboratory, the University of Massachusetts, Emory University, and Florida State University, NASA Fundamental Physics workshop in Solvang) on using magnetic fields to manipulate cells that highlighted the progress made on this task. In addition, two contributed talks were given at the APS March Meeting in Montreal.
Bibliography Type:	Description: (Last Updated: 04/16/2007)
Articles in Peer-reviewed Journals	Guevorkian, K.; Valles, J. M. "Varying the Effective Buoyancy of Cells using Magnetic Forces" Applied Physics Letters , Jun-2004
Dissertations and Theses	Wurzel, S. E. "Prospects for Studying Negative Gravitaxis in Paramecium using Magnetic Field Gradient Levitation" Brown University, Physics Department Senior Thesis May-2003
Dissertations and Theses	Mihalusova, M. "Preliminary Analysis of Microtubule Pattern Formation" Brown University, Department of Molecular Biology, Cell Biology, and Biochemistry, Senior Thesis May-2004